

F715 — Casey’s dynamical-selection idea is the NATIVE Big-Bang-unfreeze frame (already partly built), and the “stability = observability” unification is a real, buildable linear-algebra claim on the one domain. Corpus-reconnection first (before treating it as new): **BST_Big_Bang_Unfreeze** already states selection as “*selection by consistency... the only option that works... self-sustaining... thermodynamically stable and can persist,*” and thesis-topic-71 is already “prove the $SO(2)$ unfreezing is the unique self-sustaining symmetry breaking of $SO(7)$.” So “the geometries competed at the Big Bang and D_{IV}^5 was the stable one” is not a bolted-on hypothesis — it is BST’s own origin picture (the name *Bubble Spacetime* = bubble nucleation), with the stability computation already scoped. The new move is the UNIFICATION: the *dynamical* stability criterion and the *logical* selectors (color $a=3 \rightarrow D_{IV}^5$, 3+1-Lorentzian $\rightarrow$ type IV, E7-makes-no-spacetime, rank-1-disk-degenerate) are very likely one principle from two sides — a geometry that fails a logical selector is precisely one that loses the dynamical competition. If that holds, it takes the forcing off the observer entirely (no anthropics — “ D_{IV}^5 because it is the stable vacuum,” not “because observers require it”). The concrete realization is the same linear algebra as the generation count: a fluctuation (Hessian) operator over the space of Cartan geometries, read for its eigenvalue SIGNATURE (positive spectrum = stable; a negative mode = decays) — instead of a form’s kernel. Tier: framework HYPOTHESIS, corpus-grounded (topic-71), buildable; the payoff (stability=observability) is falsifiable *on the geometry side* via the fluctuation signature; the CMB stays a TARGET, never a datum (Casey: “the CMB is like the bible — you can quote it to say anything”).

Lyra, Mon 2026-07-27 15:24 EDT (full day). Continuing on the fluctuation lane (my offered wheelhouse; complements Keeper’s scoping, does not duplicate). Corpus-reconnected: it’s the Big-Bang-unfreeze/topic-71 frame, native. The unification is the deep claim; the fluctuation signature is the computation; the CMB is held at target per Casey.

What’s already in the corpus (so this isn’t a fresh framework)

- Selection-by-stability is native (BST_Big_Bang_Unfreeze): “the $SO(2)$ unfreezing isn’t chosen — it’s the only option that works... only the $SO(2)$ unfreezing is self-sustaining — it produces a configuration that is thermodynamically stable and can persist.” If any other generator unfroze first, “the system would return to the fully symmetric state.”
- Thesis-topic-71 is the stability theorem, already posed: “Prove that the $SO(2)$ unfreezing is the unique self-sustaining symmetry breaking of $SO(7)$ — that no other single-generator unfreezing produces a thermodynamically stable spatial phase with well-defined physics.”
- The descent is matrix-verified (Casey #14): $SO(7)=(7,0) \rightarrow 2 \text{ boosts} \rightarrow SO(5,2) \rightarrow \text{drop} \rightarrow SO(4,2) \rightarrow \text{drop} \rightarrow SO(3,1)$ Minkowski; the non-compact boost makes time. So Casey’s “geometries competed, D_{IV}^5 won” = the Big-Bang-unfreeze made global (over the space of geometries, not just the single $SO(2)$ unfreezing of

SO(7)).

The unification claim (the deep, new part)

Dynamical stability and the logical selectors are one principle. The logical selectors each say a competitor is *inconsistent*; the dynamical picture says the same competitor is *unstable* — and these coincide: | competitor | logical-selector failure | dynamical failure (claim) | |—|—|—| | E7 (a = 8) | wrong color number; and (Selector-2) not a Lorentzian-conformal group → no spacetime | carries a decaying fluctuation mode; can't hold a stable vacuum | | rank-1 disk | no boundary-stratum structure (degenerate) | Ehrenfest-Tangherlini-type: can't bind states / no well-posed dynamics | | wrong-n type IV | wrong emergent spacetime dimension | Ehrenfest / Tegmark-1997: unstable orbits / no well-posed time | | D_IV⁵ | passes every selector | unique positive-spectrum (stable) point | If the decaying modes line up, geometry-by-geometry, with the logical failures, the two selection routes are the same principle — and the whole forcing chain becomes one stability argument, off the observer.

The linear algebra (the buildable object — Casey's frame)

"D_IV⁵ is stable" = "D_IV⁵ is a positive-spectrum critical point of an action over the space of Cartan geometries." Concretely: 1. The configuration space: the space of (bounded symmetric domain / Cartan) geometries — parametrized by (type, rank, dimension) i.e. by the root-data / the unfreezing direction of SO(7)→.... Finite-dimensional discrete + the continuous unfreezing parameters. 2. The action / free energy whose critical points are the candidate vacua (the "self-sustaining" condition of Big_Bang_Unfreeze made a functional — the heat-trace / induced-action machinery of F60–F66 is the natural candidate for it). 3. The fluctuation operator = the Hessian of that action at D_IV⁵ — a finite Hermitian matrix. Read its eigenvalue signature: all-positive ⇒ stable (a true vacuum); a negative eigenvalue ⇒ a decay direction. This is the *same machinery* as the generation Gram matrix (a contravariant Hermitian form on the one domain), read for signature instead of kernel. 4. The test (what makes it a derivation, not a story): compute the fluctuation spectrum at D_IV⁵ and at each competitor; if D_IV⁵ is the *unique* all-positive point and each competitor's negative mode *points toward* its logical-selector failure, the unification is derived, not asserted.

Discipline (held)

- Framework HYPOTHESIS, corpus-grounded (topic-71) — not claimed proved. The unification (stability = logical selectors) is the target; the fluctuation signature is the computation that would establish it.
- CMB stays a TARGET, never a datum (Casey's "bible" point — the low- ℓ anomalies are $\sim 2-3\sigma$, cosmic-variance-limited, and consistent with a dozen pre-inflationary stories; "consistent with a bubble collision" ≠ "evidence for one"). The bubble-collision scar prediction (against Feeney-Johnson-Peiris searches) is a *second-step* target that must predict a specific signal the competition can't fake — the same discipline that retired $\sin^2\theta_W = 3/13$ and scoped confinement.
- No anthropics claimed as the mechanism — the point of the unification is precisely to *remove* the observer from the forcing (a friendlier claim to a physicist than the MUH's anthropic selection). But it is a hypothesis until the fluctuation computation lands.

Tiers / handoffs

- @Keeper (the fluctuation-lane you're scoping) — corpus-reconnection: the lane is thesis-topic-71 made global (self-sustaining unfreezing → the fluctuation Hessian over the space of geometries). I can supply the linear-algebra structure: the Hessian object, its signature test, and the *unification test* (do the negative modes match the logical-selector failures?). The action functional is the open piece — the F60–F66 induced-action/heat-trace is the natural candidate; that's where I'd start. Not front-running your charter — offering the mathematical object it computes.
- @Cal — your dynamical-selection charter + the Ehrenfest-Tangherlini / Tegmark-1997 stability inputs are exactly the "dynamical failure" column above. The CMB line you held (target, not evidence) is right and I reinforced it (Casey's "bible" point). Selector-2's "type IV is the Lorentzian-conformal family" is the E7 dynamical-failure entry.

- @Casey — your dynamical idea turns out to be the theory’s own origin story, already half-written: the Big-Bang “unfreeze” note already says the surviving geometry is the *self-sustaining, stable* one, and there’s a thesis problem on the books to prove the $SO(2)$ unfreezing is the unique stable one. Your move makes it global and — this is the prize — likely *unifies* it with all our logical reasons for D_{IV}^5 : a geometry fails our checklist for the same reason it would fail to hold together dynamically. If that’s true, the universe is D_{IV}^5 because it’s the stable one, full stop — no observers required, which is a much easier thing to say to a physicist than the mathematical-universe version. And it’s your linear algebra again: “stable” means a certain matrix (the second-variation of the action) has all-positive eigenvalues, and the competitors each have a negative one pointing at exactly the flaw we already named. On the CMB — you’re right, it’s a Rorschach; I’ve pinned it as a target we could *aim* at once the dynamics predicts a specific scar, never a reading we lean on. The buildable next step is the eigenvalue computation; I’ll take the linear-algebra structure of it with Keeper.

Notes only; no toys/theorems claimed. F715: Casey’s dynamical-selection = the NATIVE Big-Bang-unfreeze frame (BST_Big_Bang_Unfreeze: “self-sustaining/thermodynamically stable”; thesis-topic-71: “ $SO(2)$ unfreezing is the unique self-sustaining symmetry breaking of $SO(7)$ ”). The UNIFICATION claim: dynamical stability = the logical selectors (E7/rank-1/wrong-n each fail both ways; D_{IV}^5 passes both) → forcing off the observer, no anthropics. The linear algebra: the fluctuation (Hessian-of-the-action) operator over the space of Cartan geometries, read for SIGNATURE (all-positive = stable; negative mode = decays) — same machinery as the generation Gram matrix, signature instead of kernel. TEST: do the negative modes match the logical-selector failures? Action functional = F60–F66 induced-action candidate (open). Tier: framework HYPOTHESIS, corpus-grounded, buildable. CMB = TARGET only (Casey’s “bible” point; $\sim 2-3\sigma$, cosmic-variance, Rorschach) — the scar prediction (Feeney–Johnson–Peiris) is a second-step target that must predict a signal the competition can’t fake. — Lyra